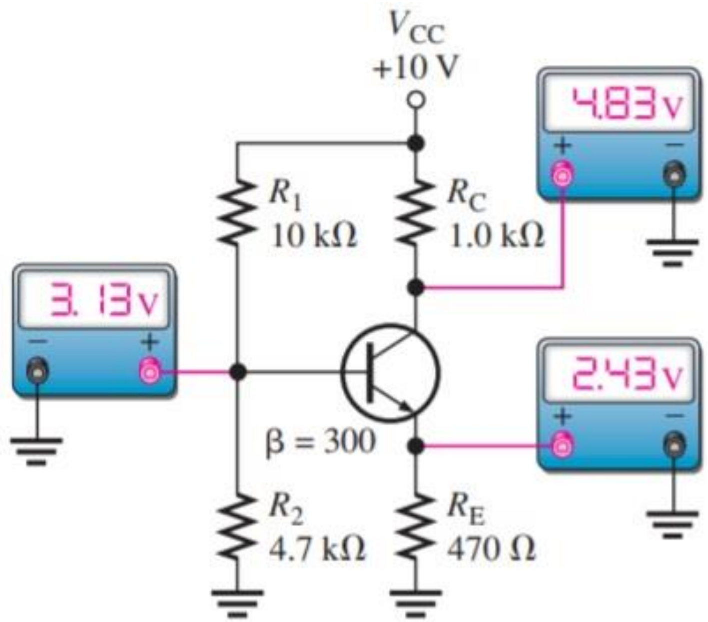


ANALOG ELETRONICS

LECTURE 2:

TRANSISTOR BIAS CIRCUITS

Instructor: E.Tarimo **Wednesday, April 28, 2021**

2020/2021

OUTLINE

- Introduction
- Application
- The DC Operating Point
- Voltage-Divider Bias
- Other Bias Methods

Introduction

- As you learned in lecture 1, a transistor must be properly biased in order to operate as an amplifier.
- DC biasing is used to establish fixed dc values for the transistor currents and voltages called the dc operating point or quiescent point (Q-point).
- In this lecture, several types of bias circuits will be discussed.
- Therefore this discussion lays the groundwork for the study of amplifiers in lecture 3, and other circuits that require proper biasing.

APPLICATION

- One of the main application of transistor biasing focuses on a system for controlling temperature in an industrial chemical process.
- This is a circuit that converts a temperature measurement to a proportional voltage that is used to adjust the temperature of a liquid in a storage tank.
- The first step is to learn all you can about transistor operation.
- You will then apply your knowledge to solve problems related to this application activities.

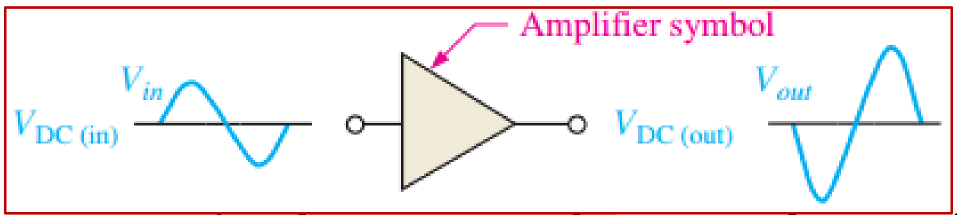
THE DC OPERATING POINT

- A dc operating point must be set so that signal variations at the input terminal are amplified and accurately reproduced at the output terminal.
- This means that, at the dc operating point, I_C and V_{CE} have specified values.
- The dc operating point is often referred to as the Q-point (quiescent point).

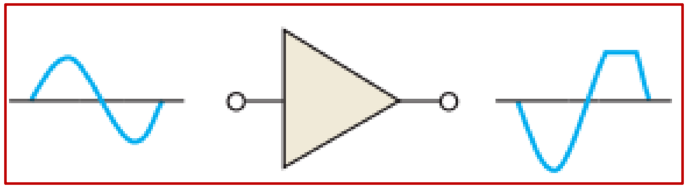
THE DC OPERATING POINT

DC Bias

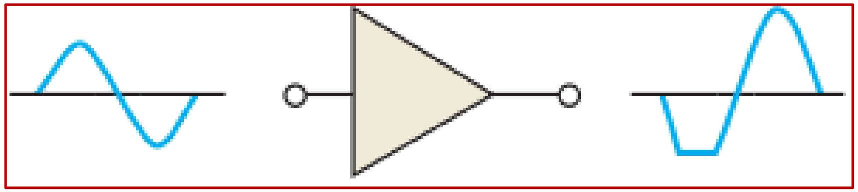
- Bias establishes the dc operating point (Q-point) for proper linear operation of an amplifier.
- If an amplifier is not biased with correct dc voltages on the input and output, it can go into saturation or cutoff when an input signal is applied.



Linear operation: larger output has same shape as input except that it is inverted



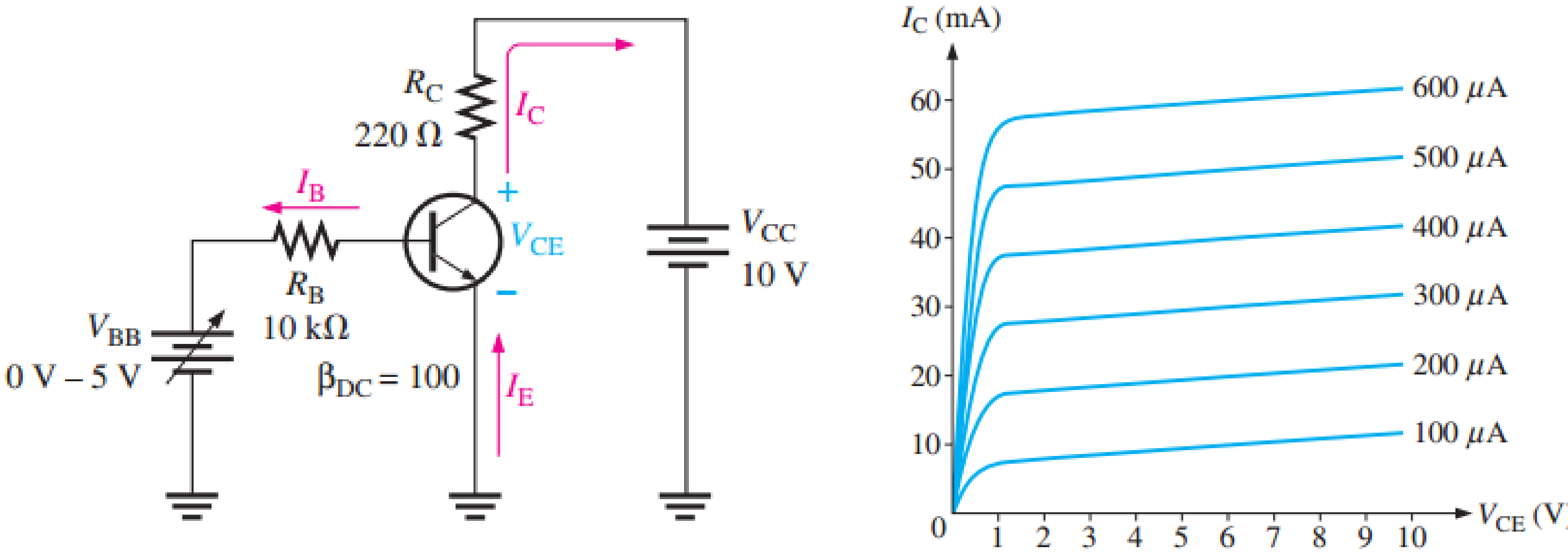
Nonlinear operation: output voltage limited (clipped) by cutoff



Nonlinear operation: output voltage limited (clipped) by saturation

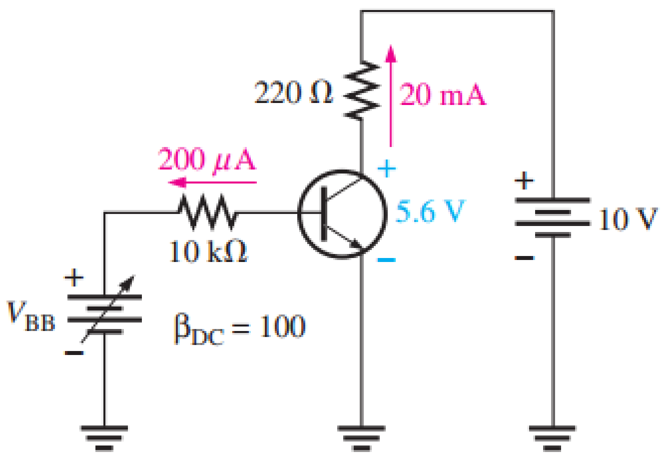
THE DC OPERATING POINT

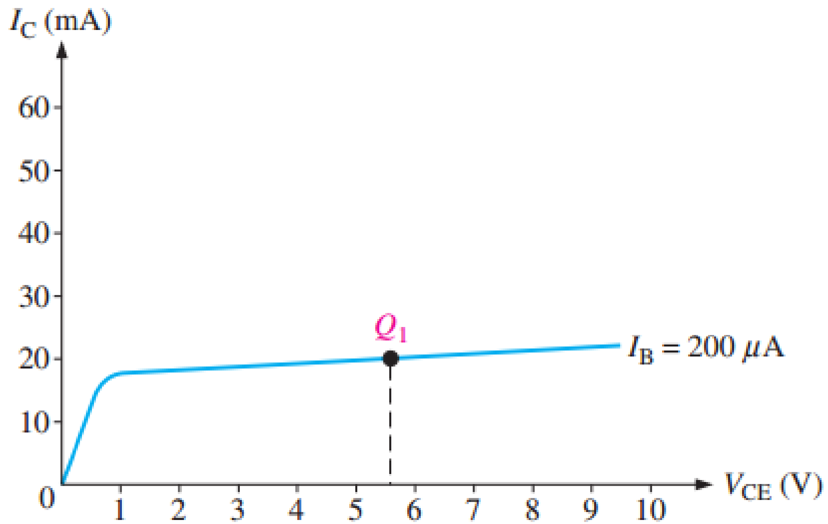
- Consider a dc-biased transistor circuit with variable bias voltage (V_{BB}) for generating the collector characteristic curves.



THE DC OPERATING POINT

- First, V_{BB} is adjusted to produce an I_B of 200 μA.
- Since $I_C = \beta_{DC} I_B$ the collector current is 20 mA

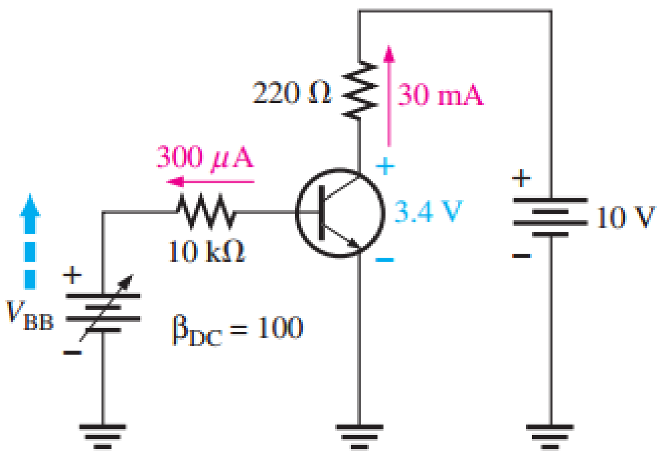


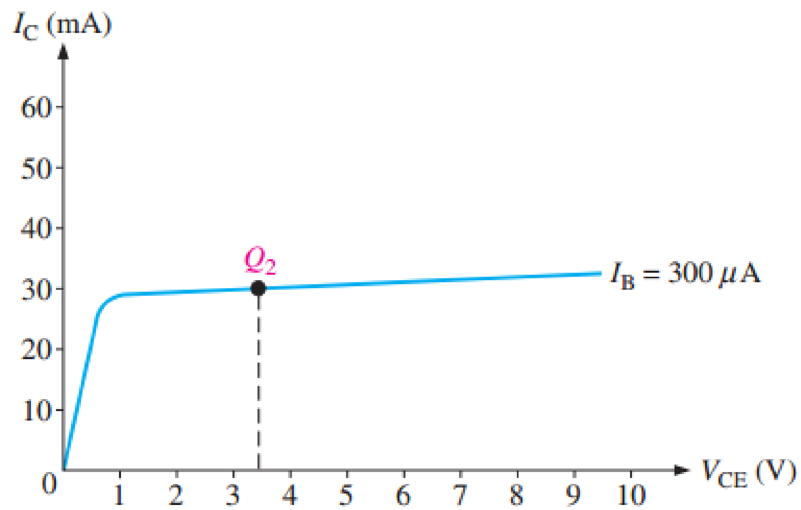


- $V_{CE} = V_{CC} - I_C R_C = 10V - 20mA\,(220\Omega) = 5.6V$
- The Q-point Q_1 is (V_{CE}, I_C) = (5.6V, 20mA)

THE DC OPERATING POINT

- Next, V_{BB} is increased to produce an I_B of 300 μA and an I_C of 30 mA.



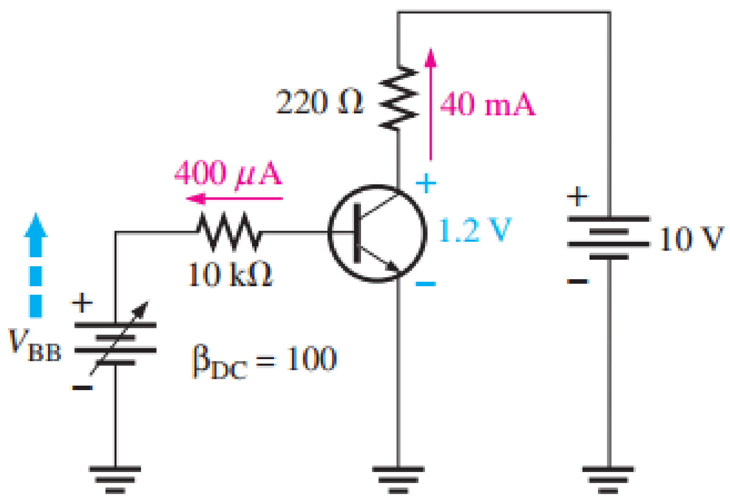


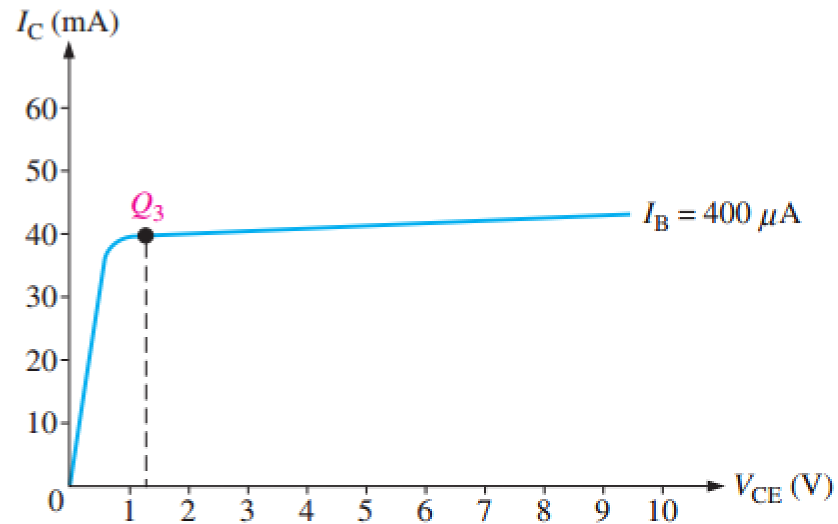
- $V_{CE} = V_{CC} - I_C R_C = 10V - 30mA\,(220\Omega) = 3.4V$

- The Q-point Q_1 is $(V_{CE}, I_C) = $ (3.4V, 30mA

THE DC OPERATING POINT

- Finally, V_{BB} is increased to give an I_B of 400 μA and an I_C of 40 mA



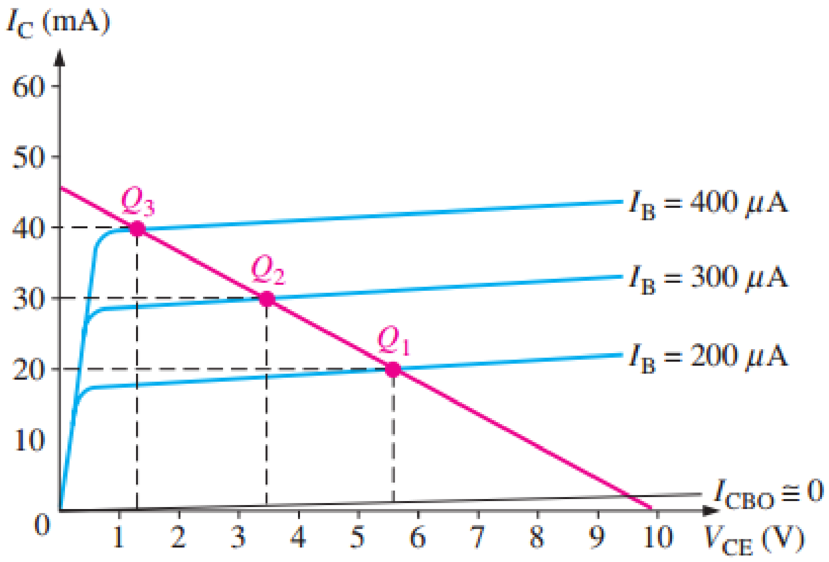


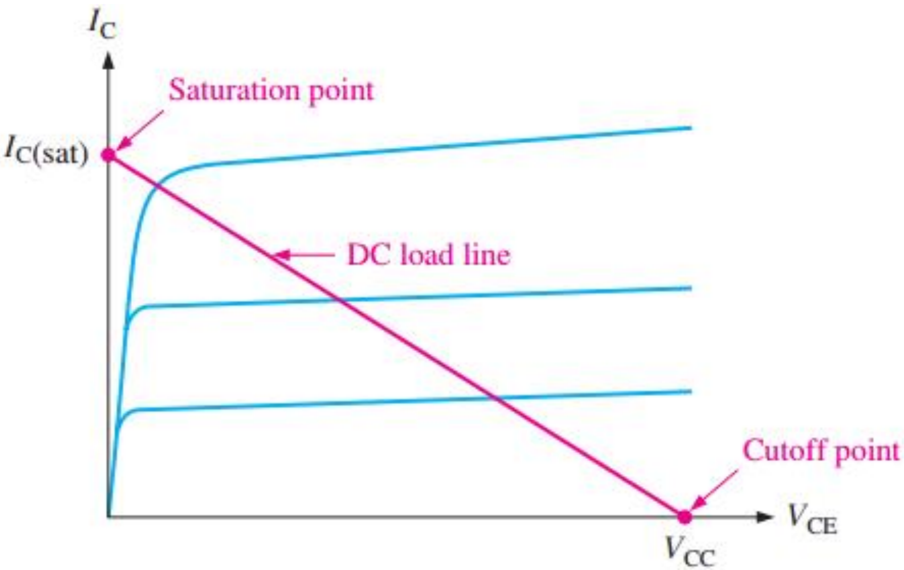
- $V_{CE} = V_{CC} - I_C R_C = 10V - 40mA\,(220\Omega) = 1.2V$
- The Q-point Q_1 is (V_{CE}, I_C) = (1.2V, 30mA

THE DC OPERATING POINT

DC Load Line

- This is a straight line drawn on the characteristic curves from the saturation value where $I_C = I_{C(sat)}$ on the y-axis to the cutoff value where $V_{CE} = V_{CC}$ on the x-axis





THE DC OPERATING POINT

- From a dc-biased transistor circuit;

$$V_{CC} = I_C R_C + V_{CE}$$

- Now the equation for I_C is;

$$I_C = \frac{V_{CC} - V_{CE}}{R_C} = \frac{V_{CC}}{R_C} - \frac{V_{CE}}{R_C}$$

$$I_C = -\frac{V_{CE}}{R_C} + \frac{V_{CC}}{R_C} = -\left(\frac{1}{R_C}\right)V_{CE} + \frac{V_{CC}}{R_C}$$

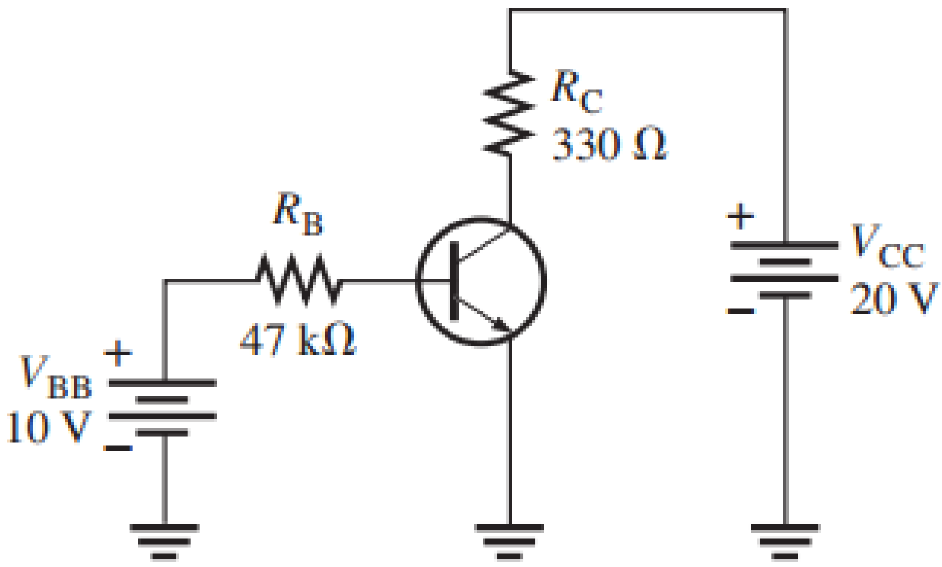
$$I_C = -\left(\frac{1}{R_C}\right)V_{CE} + \frac{V_{CC}}{R_C}$$

- This is the equation of a straight line with a slope of $\boldsymbol{-1/R_C}$, an x intercept of $\boldsymbol{V_{CE} = V_{CC}}$, and a y intercept of V_{CC}/R_C, which is $\boldsymbol{I_{C(sat)}}$

THE DC OPERATING POINT

Example 1

- Determine the Q-point for the circuit in figure and draw the dc load line. Assume $\beta_{DC} = 200$



THE DC OPERATING POINT

Solution

- The Q-point is defined by the values of I_C and V_{CE}.

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{10V - 0.7V}{47\ k\Omega} = 198\ \mu A$$

$$I_C = \beta_{DC} I_B = 200(198\ \mu A) = \mathbf{39.6\ mA}$$

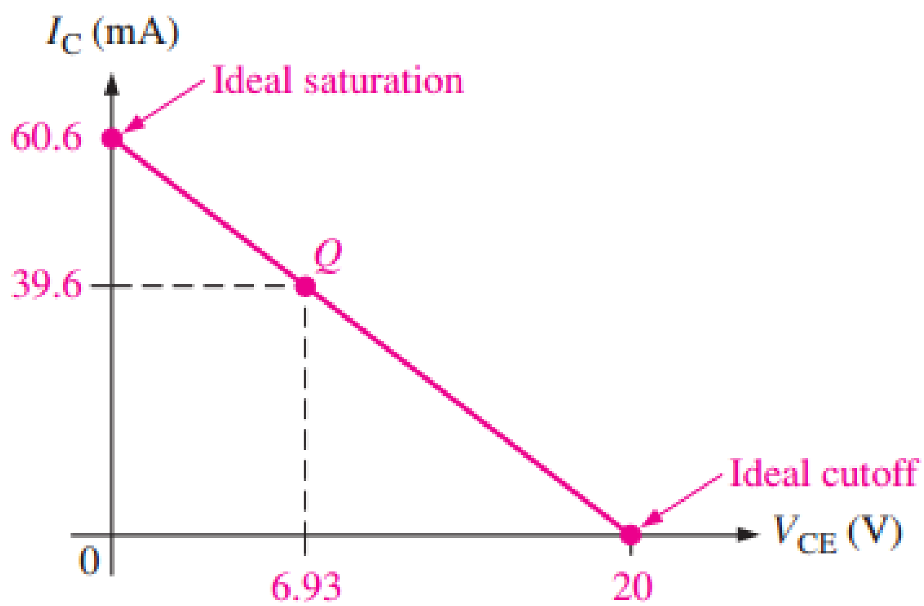
$$V_{CE} = V_{CC} - I_C R_C = 10\ V - 13.7\ V = \mathbf{6.93 V}$$

- The Q-point is at $I_C = 39.6\ mA$ and at $V_{CE} = 6.93\ V$.

THE DC OPERATING POINT

- Since at cutoff; $I_{C(cutoff)} = 0$, then $V_{CE} = V_{CC} = 20\ V$
- Now at saturation; $I_{C(sat)}$ can be calculated as;

$$I_{C(sat)} = \frac{V_{CC}}{R_C} = \frac{20\ V}{300\ \Omega} = 60.6\ \mathrm{m}A$$

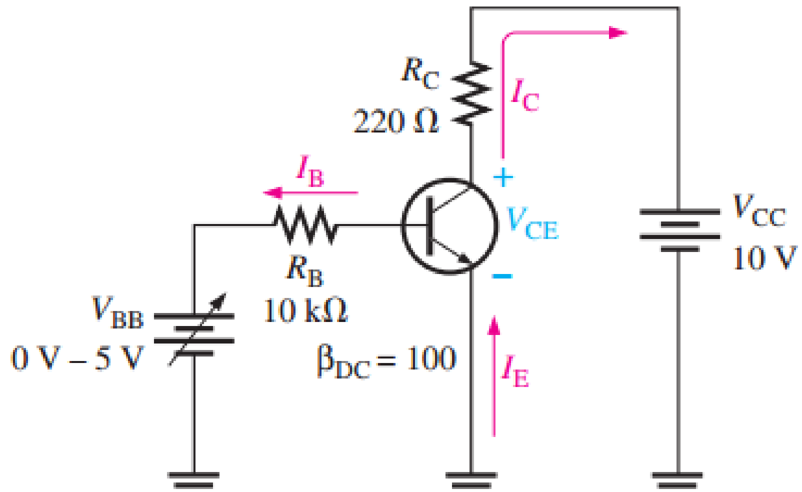


THE DC OPERATING POINT

Review question

1. Determine the Q-point for the circuit in figure of example 1 and draw the dc load line for the following circuit values: $\beta_{DC} = 100$, $R_C = 1\ k\Omega$ and $V_{CC} = 24\ V$.

2. What is the Q-point for a biased transistor as in Figure below. $I_B = 150\mu A$, $\beta_{DC} = 75$, $V_{CC} = 18\ V$ and $R_C = 1\ k\Omega$

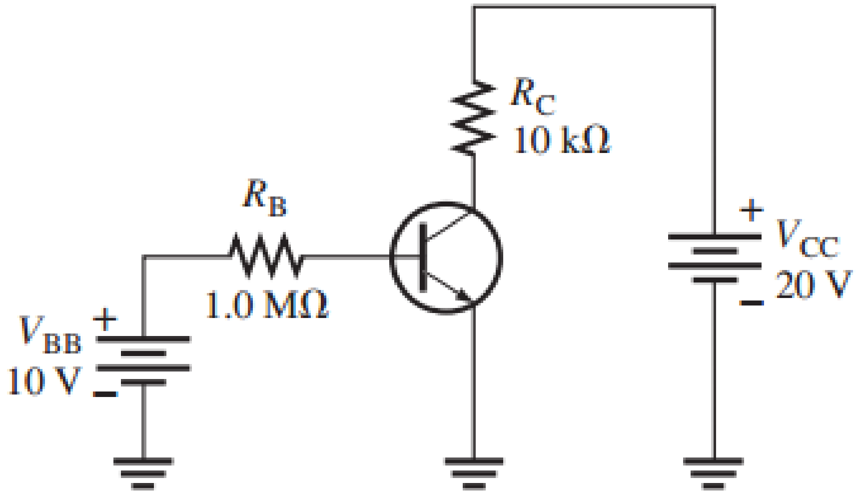


THE DC OPERATING POINT

Review question

3. What is the saturation value of collector current and the cutoff value of V_{CE} in Problem 2? Draw dc load line.

4. Determine the intercept points of the dc load line on the vertical and horizontal axes of the collector-characteristic curves for the circuit in Figure below.



THE DC OPERATING POINT

Review question

5. Assume that you wish to bias the transistor in Figure of problem 4 with $I_B = 20\ \mu A$ To what voltage must you change the V_{BB} supply? What are I_C and V_{CE} at the Q-point, given that $\beta_{DC} = 50$?

6. From the collector characteristic curves and the dc load line in Figure given in next page, determine the following:

(a) Collector saturation current

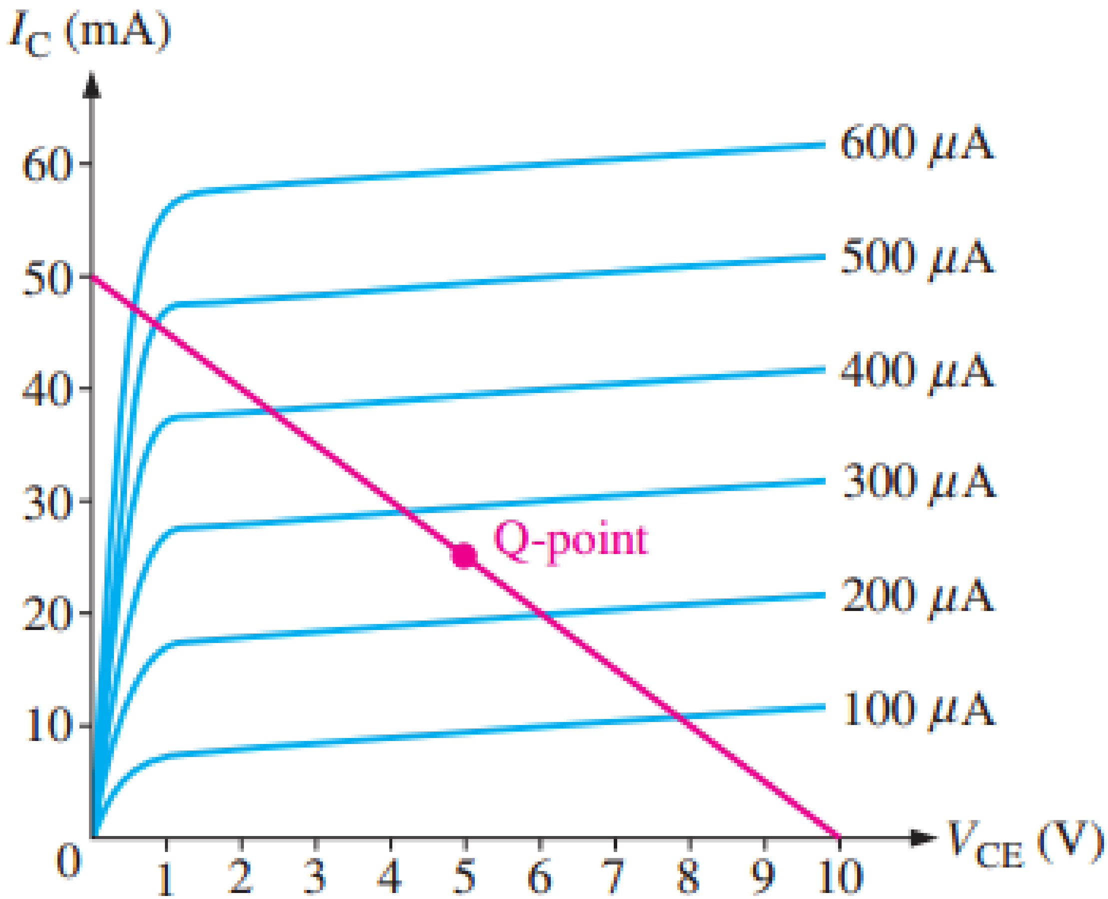
(b) V_{CE} at cutoff

(c) Q-point values of I_B, I_C, and V_{CE}

THE DC OPERATING POINT

Review question

6.



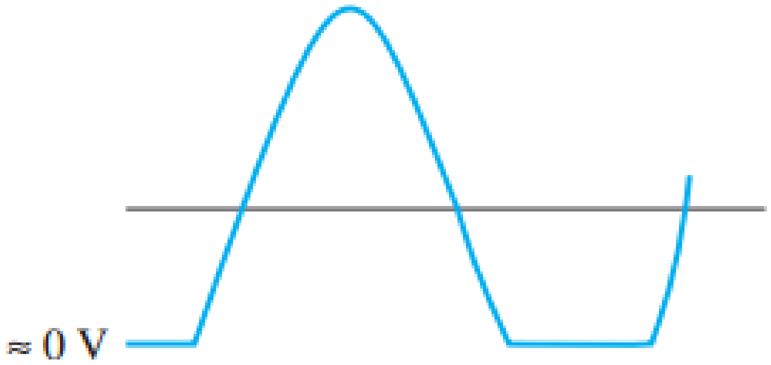
THE DC OPERATING POINT

Review question

7. (i) At what point on the load line does saturation occur? (ii) At what point does cutoff occur?

8. Define Q-point

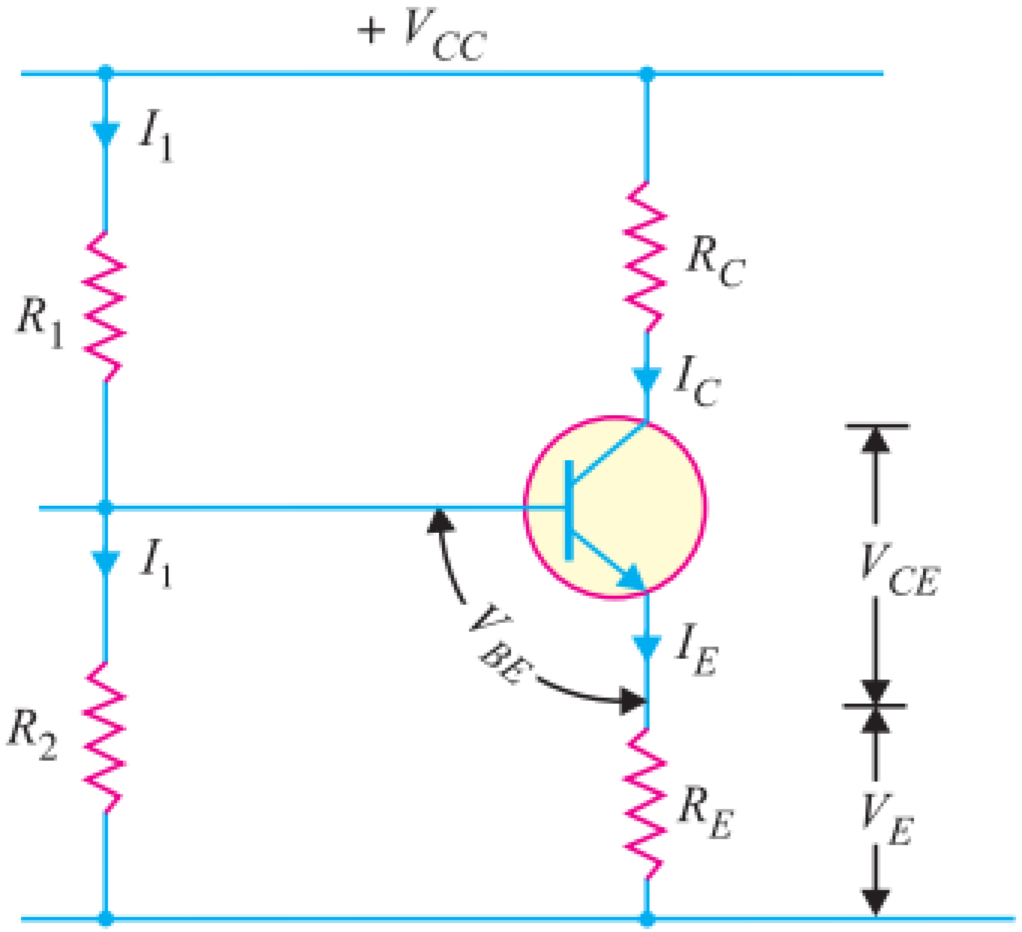
9. The output (collector voltage) of a biased transistor amplifier is shown in the Figure below. Is the transistor biased too close to cutoff or too close to saturation?



Voltage-Divider Bias

- In this method, two resistances R_1 and R_2 are connected across the supply voltage V_{CC} and provide biasing.
- The emitter resistance R_E provides stabilization.
- The name "voltage divider" comes from the voltage divider formed by R_1 and R_2.
- The voltage drop across R_2 forward biases the base-emitter junction.
- This causes the base current and hence collector current flow in the zero signal conditions.

VOLTAGE-DIVIDER BIAS



Voltage-Divider Bias

Collector current I_C

- Suppose that the current flowing through resistance R_1 is I_1.
- As base current I_B is very small, therefore, it can be assumed with reasonable accuracy that current flowing through R_2 is also I_1.

$$I_1 = \frac{V_{CC}}{R_1 + R_2}$$

$\therefore$ Voltage across resistance R_2 is

$$V_2 = \left(\frac{V_{CC}}{R_1 + R_2}\right) R_2$$

Voltage-Divider Bias

- Applying Kirchhoff 's voltage law to the base circuit;

$$V_2 = V_{BE} + V_E$$

$$V_2 = V_{BE} + I_E R_E$$

$$I_E = \frac{V_2 - V_{BE}}{R_E}$$

Since $\quad I_C \cong I_E$

$$I_C = \frac{V_2 - V_{BE}}{R_E}$$

Voltage-Divider Bias

- It is clear from expression that I_C does not at all depend upon β.
- Though I_C depends upon V_{BE} but in practice $V_2 >> V_{BE}$ so that I_C is practically independent of V_{BE}.
- Thus I_C in this circuit is almost independent of transistor parameters and hence good stabilization is ensured.
- It is due to this reason that potential divider bias has become universal method for providing transistor biasing.

Voltage-Divider Bias

Collector-emitter voltage V_{CE}

- Applying Kirchhoff 's voltage law to the collector side,

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E$$

Since $I_C \cong I_E$

$$V_{CC} = I_C R_C + V_{CE} + I_C R_E$$

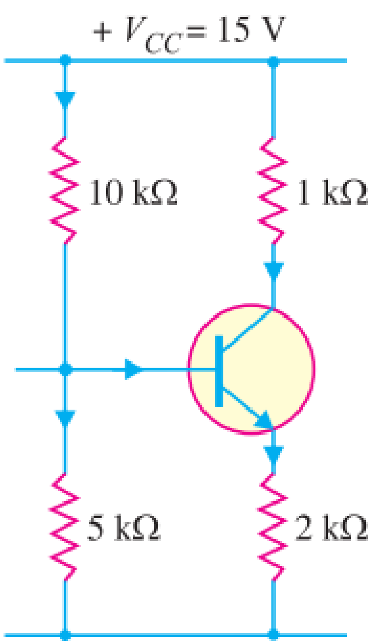
$$V_{CC} = I_C (R_C + R_E) + V_{CE}$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

Voltage-Divider Bias

Example 2

- The figure below shows the voltage divider bias method. Draw the d.c load line and determine the operating point. Assume the transistor to be of silicon.



VOLTAGE-DIVIDER BIAS

Solution

d.c load line

- The collector-emitter voltage V_{CE} is given by :

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

- When $I_C = 0$,

$$V_{CE} = V_{CC} = 15\ V.$$

- When $V_{CE} = 0$,

$$I_C = \frac{V_{CC}}{R_C + R_E} = \frac{15\ V}{(1+2)\ k\Omega} = 5\ \mathrm{m}A$$

Voltage-Divider Bias

Solution

Operating point

- For silicon transistor,

$$V_{BE} = 0.7\ V$$

- Voltage across 5 kΩ is;

$$V_2 = \left(\frac{V_{CC}}{R_1 + R_2}\right) R_2 = \left(\frac{15}{10+5}\right) 5 = 5\ V$$

∴ Emitter current,

$$I_E = \frac{V_2 - \ V_{BE}}{R_E} = \frac{5 - 0.7}{2\ k} = \frac{4.3}{2\ k} = 2.15\ \mathrm{m}A$$

∴ Collector current is $I_C \ \cong \ I_E = 2.15\ \mathrm{m}A$

Voltage-Divider Bias

Solution

Collector-emitter voltage,

$$V_{CE} = V_{CC} - I_C(R_C + R_E)$$

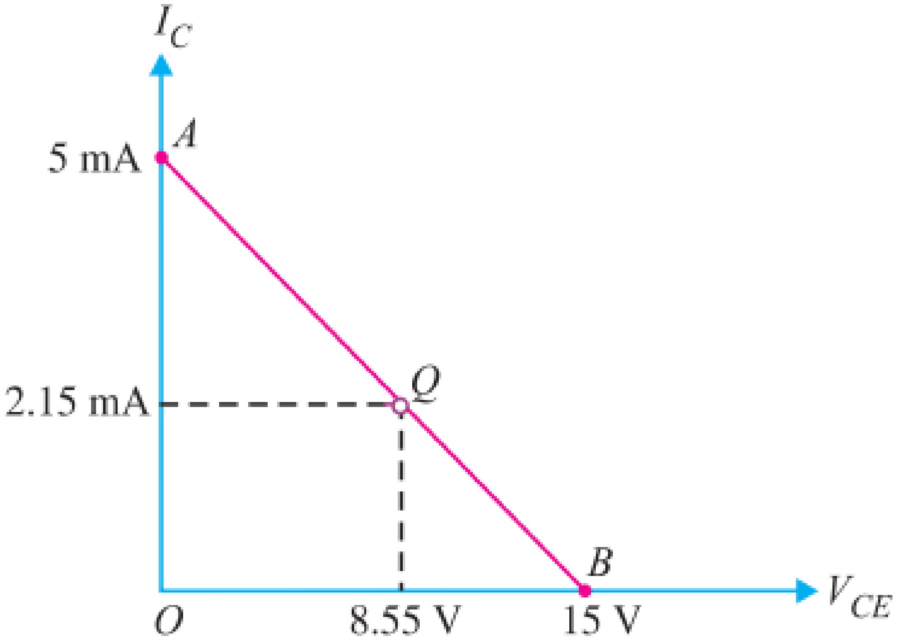
$$V_{CE} = 15\,V - 2.15\,mA(3k\Omega)$$

$$V_{CE} = 15\,V - 6.65\,V$$

$$V_{CE} = 8.55\text{ V}$$

∴ Operating point is

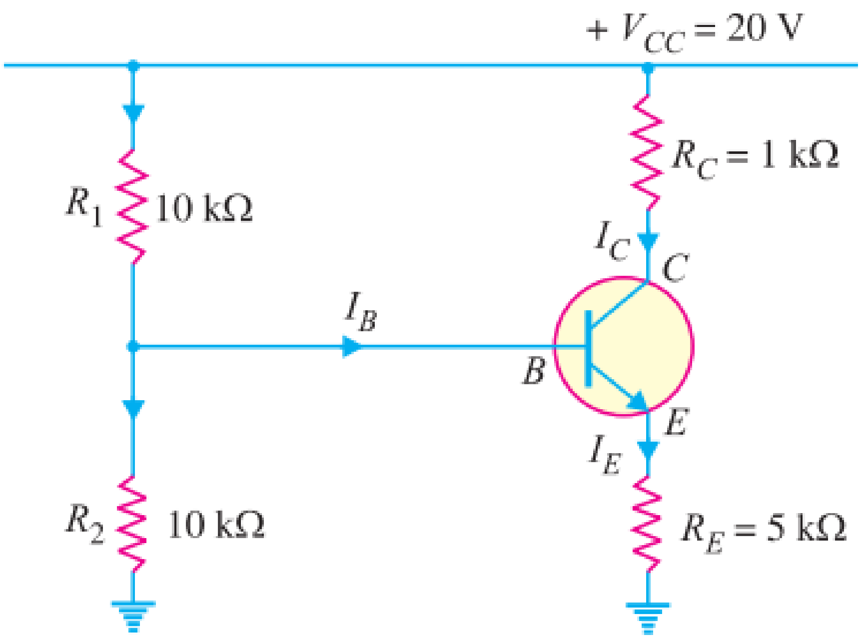
8.55 V, **2.15 mA**.



VOLTAGE-DIVIDER BIAS

Review question

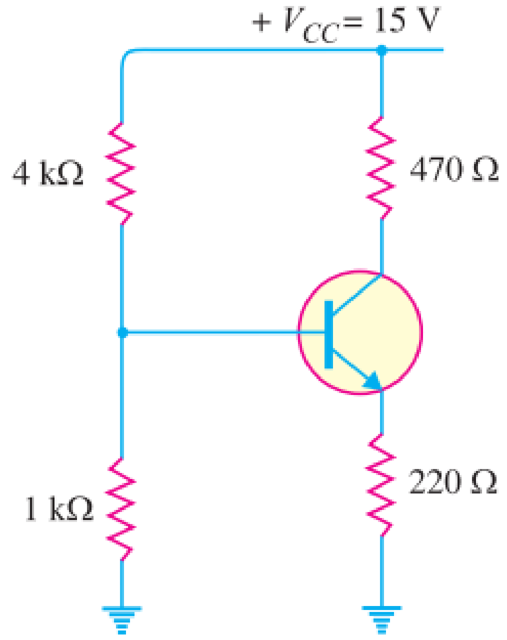
1. Calculate the emitter current in the voltage divider circuit shown in the below. Also find the value of V_{CE} and collector potential V_C . (*Answer*: $I_E = 2mA, V_{CE} = 8V,$ $V_C = 18V$)



VOLTAGE-DIVIDER BIAS

Review question

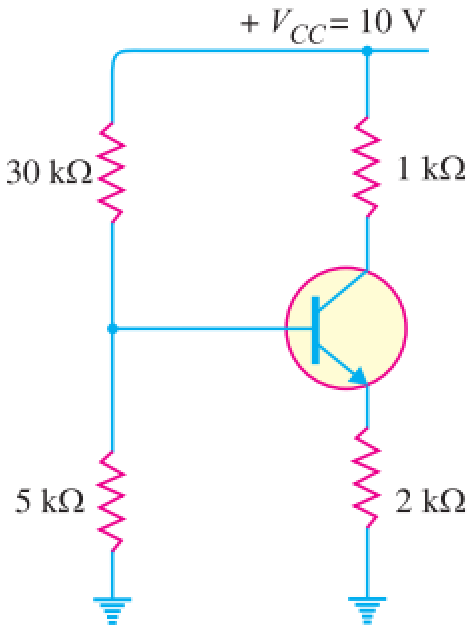
2. In the transistor circuit shown in fig below, find the operating point and draw dc load line. Assume the transistor to be of silicon. (*Answer*: $I_C = 10.5\,mA, V_{CE} = 7.75\,V$)



Voltage-Divider Bias

Review question

3. In a transistor circuit shown in Fig below, find the operating point and draw dc load line. Assume silicon transistor is used. (*Answer*: $I_C = 0.365\ mA,\ V_{CE} = 8.9\ V$)



VOLTAGE-DIVIDER BIAS

Review question

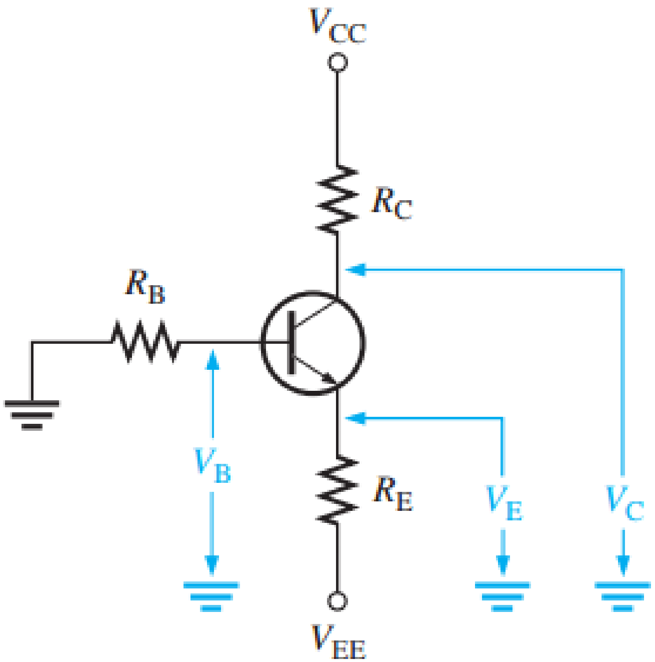
4. Find the value of I_C for potential divider method if $V_{CC} = 9\,V$, $R_E = 1k\Omega$, $R_1 = 39k\Omega$, $R_2 = 10k\Omega$, $R_C = 2.7k\Omega$, $V_{BE} = 0.15\,V$ and $\beta = 90$. ($Answer$: $I_C = 1.5\,mA$)

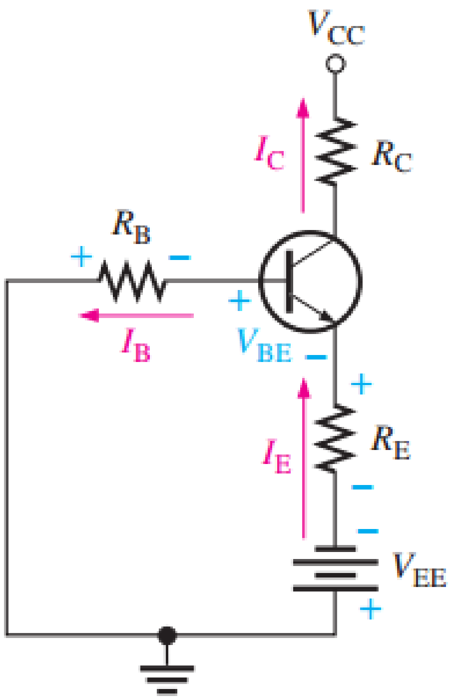
Other Bias Methods

- In this section, four additional methods for dc biasing a transistor circuit are introduced.
- Although these methods are not as common as voltage-divider bias,
- But you should be able to recognize them when you see them and understand the basic differences.

Other Bias Methods

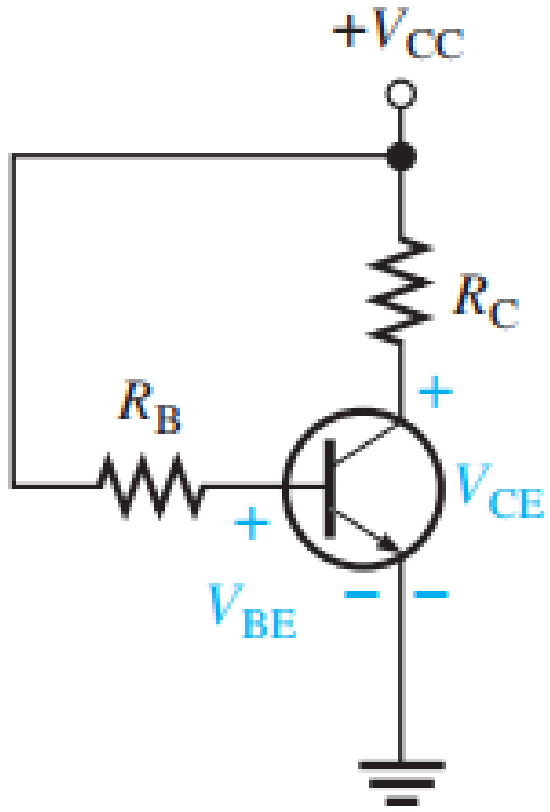
- **Emitter Bias**





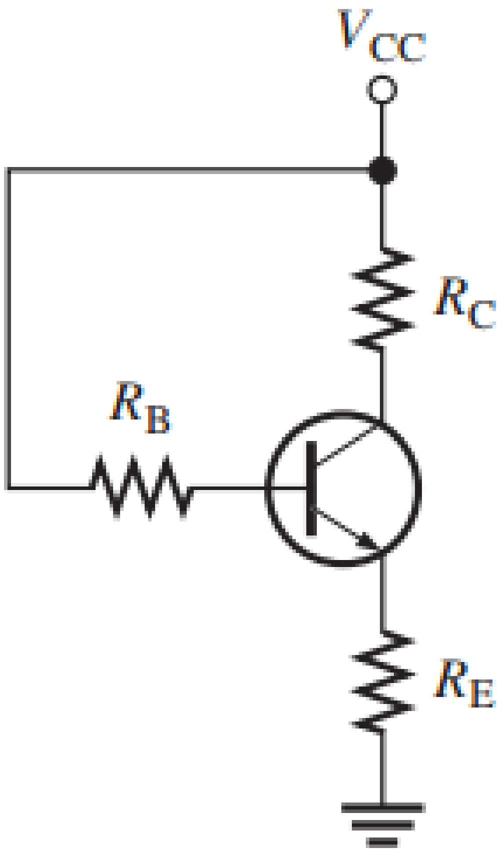
Other Bias Methods

- **Base Bias**



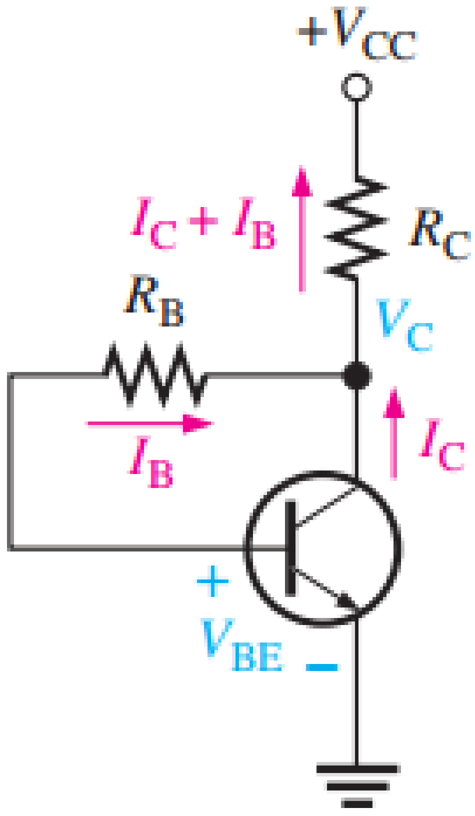
Other Bias Methods

- **Emitter-Feedback Bias**



Other Bias Methods

- **Collector-Feedback Bias**



NEXT SESSION…TRANSISTOR AMPLIFIERS